ANALYTICAL CHEMISTRY

CHE 2103

1 Credit (15 lecture hours)

Pre-requisites:- All chemistry courses of level 1

Course aims:

- To obtain a strong foundation in chemical analysis
- To describe concepts and principles used in analytical chemistry
- To develop skills to solve problems in quantitative chemical analysis based on classical techniques.
- Introduce the assessment of errors and of the accuracy of experiments.

Course aims: ………..

- Introduce data handling.
- Analysis and fitting to calculate results
- Introduce some basic principles of measurement in physical, organic and analytical chemistry.
- Provide the students with analytical and numerical tools to solve chemistry related problems.

Course ILOs:

- On successful completion of the course students should be able to:-
- Carry out propagation of uncertainty in calculations and describe the type of experimental errors.
- Apply several common statistical tests to experimental data.
- Construct a linear regression line from experimental data

Course ILOs:………

- Demonstrate chemical concentrations in unit of molarity, molality, percent composition, part per million and parts per billion.
- Determine which technique is to be used based on an understanding of experimental parameters.
- Recognize and use the different types of titrimetric analysis methods (acid-base , complexation, redox and precipitation)

Course ILOs:………

- Select appropriate buffer systems for a given pH requirement and describe step by step procedure for buffer preparation.
- Describe and apply gravimetric methods of analysis.
- Demonstrate accurate manipulation and critical thinking.

Course Description

- Measurements and errors in chemical analysis (05 h)
- Significant figures, statistical treatment of analytical data, t-test, F-test and Q-test, classification of errors and means to minimize errors; Sampling procedures, sample population, significance of representative sampling, working curve, blank solution , standard addition technique, curve fitting, graphical analysis, Quality Control/Quality assurance.

Course Description……….

- Classical Methods (10 h)

- Acid base titrations: The requirements and the feasibility of acid base titrations, calculation of the equivalence point pH value and construction of pH curves; Theory of indicators and apparent indicator constant concept; Titration of mono- and poly functional acid and bases; Titration in non-aqueous solvents and indicators for non-aqueous titration.

Course Description………..

- Buffer solution: Derivation of the Henderson-Hasselbalch equation; Calculation of pH and preparation of buffer solutions, buffer capacity; Complexoetric titrations: Concept of complexation and chelation; Coordination number, stability of complexes, factors affecting stability constants, selectivity of complexometric titrations, types of EDTA titrations; metallochromic indicators and analysis of mixtures.

Course Description……….

- Gravimetry: Properties of precipitates and precipitating regent, particle size and filterability of precipitates, colloidal and crystalline precipitates, co-precipitation and post-precipitation drying and ignition of precipitates, principles of gravimetric estimation of chloride, phosphate, oxalate, zinc , iron, aluminum and magnesium.

Teaching Methods:

- Lectures
- Assignments
- Tutorials and quizzes

Course Evaluation

	Contribution to course Grade (%)
Continuous assessments	10
Mid semester examination	20
End semester examination theory	70

References:

- D.A.Skoog, D.M.West and F.J.holler Fundamentals of analytical chemistry, 8th edition.
- Gary D. Christian, Analytical Chemistry, 6th edition.

- Measurements and errors in chemical analysis (05 h)
- Significant figures, statistical treatment of analytical data, t-test, F-test and Q-test, classification of errors and means to minimize errors; Sampling procedures, sample population, significance of representative sampling, working curve, blank solution , standard addition technique, curve fitting, graphical analysis, Quality Control/Quality assurance.

Significant Figures

Significant Figures (digits)

- = reliable figures obtained by measurement
- = all digits known with certainty plus one estimated digit

Taking the measurement

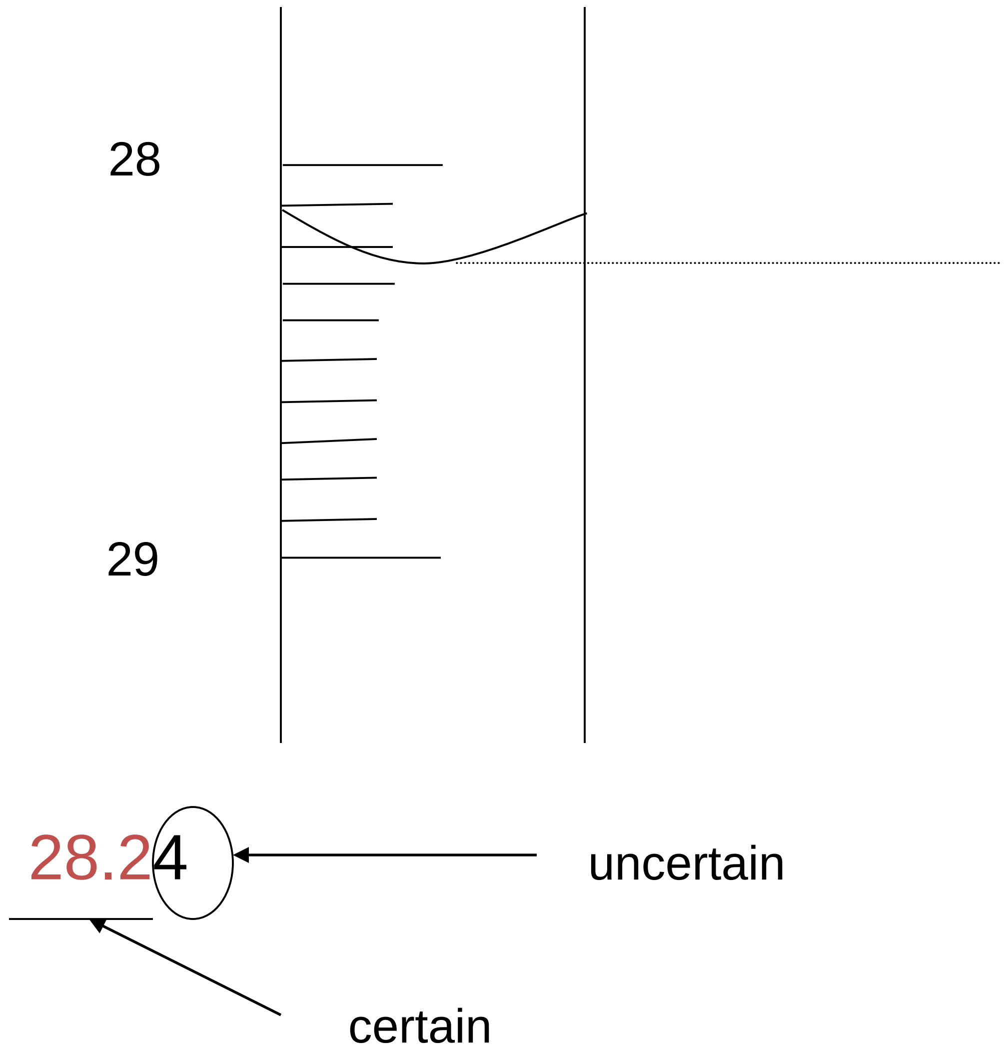
- Is always some uncertainty
- Because of the limits of the instrument you are using

Significant Figures

(in numerical computations)

- We often indicate the probable uncertainty associated with an experimental measurement by rounding the result which contains only significant figures.
- Significant figures in a number are all of the digit known certainty plus the first uncertain digit.

- For eg: consider the burette reading as



Rules for determining the number of significant figures.

- The digit zero may or may not be a significant figure, depending upon its function in the number.
- For eg:- In a buret reading of <u>10.06</u>
- Both zeros are measured numbers and are therefore significant.

1. Disregard all initial zeros:

   30.24 mL and 0.03024 L are same number of significant figures number of significant figures = 4. in 0.03024 L is used to place the decimal.

2. Disregard all final zeros unless they follow a decimal point.

   for example: 2.0 L and 2000 mL

Not significant

uncertainty

Significant no. = 2

- 2000 mL can be written as 2.0×10^3 mL
- 3. All remaining digits, including zeros between nonzero digits are significant. For example 4.02 has number of significant figures 3).
- 4. Any digit that is non zero is significant. For example; 837 cm has three significant figures and 1.235 kg has four significant figures.
- 5. For number that that do not contain decimal points, the trailing zeros may or may not be significant. 400 cm may have one significant figure as 4×10^2 cm or two significant figures as 4.0×10^2 cm or three significant figures as 4.00×10^2 cm .

Significant figures in numerical calculations

- 1. In addition and subtraction:
- For addition and subtraction, the number of significant figures to the right of the decimal point in the final sum or difference is determined by the lowest number of significant figures to the right of the decimal point in any of the original numbers.

- For eg (1): $3.4 + 0.020 + 7.31 = 10.730$

After the decimal one two two

The answer is **10.7**

Only one significant figure after the decimal

- Eg (2): 89.332 + 1.1 = 90.432

After the decimal three one

The answer is 90.4

The answer has only one significant figure after the decimal

Eg: $2.097 - 0.12 = 1.977$

Round off = 1.98

After the decimal three two

The answer is **1.98**

The answer has only two significant figures after the decimal

Round off a number

- To round off a number at a certain point we simply drop the digits that follow if the first of them is less than 5.
- For eg: 8.724 = 8.72 (if we want two figures after the decimal point)

(4 is < 5 , then it can drop)

8.7245 = 8.72 (if we want two figures after the decimal point)

- Always round the computed results of chemical analysis in an appropriate way.
- Consider the following replicate results: 61.60, 61.46, 61.55 and 61.61.
- The mean of the data is :
- (61.60 + 61.46 + 61.55 + 61.61) / 4 = 61.555.
- How to round 61.555 to two significant figures;
- 61.56 or 61.55

- Correct one is 61.56
- In rounding a number ending in 5 , always round so that the result ends with an even number .
- i.e., 0.635 = 0.64
- 0.625 = 0.62

- When adding and subtracting numbers in scientific notation → express the numbers to the same power of 10 in the result.
- For eg:
- $2.432 \times 10^6 + 6.512 \times 10^4 - 1.227 \times 10^5 =$
- $2.432\times10^6 + 0.06512\times10^6 - 0.1227\times10^6 =$

After decimal

three

four

four

$= 2.37442 \times 10^6$

Answer can have only three significant figures $= 2.374 \times 10^6$

three

Multiplication and division

- The number of significant figures in the final product or quotient is determined by the original number that has the smallest number of significant figure
- Eg: 2.8 x 4.5039 = 12.61092

Eg: $2.8 \times 4.5039 = 12.61092$

Round off = 13

No. of significant figures two five

The answer is 13

The answer has only two significant figures

Eg: 6.85 / 112.04 = 0.0611388789

Round off = 0.0611

No. of significant figures three five

The answer is 0.0611

The answer has only three significant

Eg: 24 x 4.52 / 100.0 = 1.0848

Round off = 1.08

No. of significant figures	---	three	four

The answer is **1.08**

The answer has only three significant

Eg: 24 x 4.02 / 100.0 = 0.965

No. of significant figures --- three four

The answer is 0.965

The answer has only three significant figures

- Exact numbers obtained from definition or by counting numbers of objects can be considered to have an infinite number of significant figures.
- Eg: If an object has the mass 0.2786 g, then the mass of 8 such object is =
- 0.2786 x 8 = 2.229 g

0.2786	8	2.229 g	
four	----	four	← Significant figures

- Eg: (6.64 cm + 6.68 cm) / 2 = 6.66 cm

Number 2 = 2.000000000…….

Significant figure after decimal = two

Logarithm and antilogarithms

- A logarithm is composed of two parts.
- log (2.4×10^5)
- A whole number, <u>the characteristic</u> plus a decimal fraction, <u>the mantissa.</u>
- The characteristic is a function of the position of the decimal in the number,
- Whose logarithm is being determined and therefore is not a significant figures.
- Mantissa is the same regardless of the position of the decimal,
- All the digits are considered significant.

- Rules are:
- Rule (1). In a logarithm of a number, keep as many digits to the right of the decimal point as there are significant figures in the original number.
- Eg: log $\underline{4.000} \times 10^{-5}$ = - 4.$\underline{3979}$400

- 4.3979

Significant figures

$\log(9.57 \times 10^4) =$

ans : 4.981

- Rule (2):- In an antilogarithm of a number, keep as many digits as there are digits to the right of the decimal point in the original number.
- antilog 12.5 = 3.1262277 x 10^{12}
- the answer is 3 x 10^{12}

Questions (for 10 minutes)

- 1) $11254.1 + 0.1983 =$
- 2) $66.59 – 3.113 =$
- 3) $8.16 \times 5.1355 =$
- 4) $0.0154 / 883 =$
- 5) $2.64 \times 10^3 + 3.27 \times 10^2 =$
- 6) $-\log(2.0 \times 10^{-3}) = 3 – 0.3010 = ?$

answers

- 1) 11254.1 + 0.1983 =
- answer = 11254.2983
- = 11254.3
- 2) 66.59 – 3.113 =
- answer = 63.477
- = 63.48

- 3). 8.16 x 5.1355 =
- answer = 41.90568
- = 41.9
- 4). 0.0154 / 883 =
- answer = 0.0000174405436
- = 0.0000174 0r
- = 1.74×10^{-5}

- 5) $2.64 \times 10^3 + 3.27 \times 10^2$ =
- answer = $2.64 \times 10^3 + 0.327 \times 10^3$
  $= 2.967 \times 10^3$
- $= 2.97 \times 10^3$
- 6) $-\log (2.0 \times 10^{-3}) = 3 - 0.30) = 2.70$

Determining Significant Figures

- State the number of significant figures in the following measurements:

2005 cm	0.050 cm
25.000 g	0.0280 g
25.0 ml	50.00 ml
0.25 s	1000 s
0.00250 mol	1000. mol

Answer

4	2
2	3
3	4
2	-
3	4

When both addition/subtraction and multiplication/division appear in the same problem

- In addition/subtraction the number of significant digits is limited by the value of greatest uncertainty.
- In multiplication/division, the number of significant digits is limited by the value with the fewest significant digits.
- Since the rules are different for each type of operation, when they both occur in the same problem,
  - complete the first operation and establish the correct number of significant digits.
  - Then proceed with the second and set the final answer according to the correct number of significant digits based on that operation

$(1.245 + 6.34 + 8.179)/7.5$

- Add

$$1.245 + 6.34 + 8.179 \quad = 15.764$$

- $= 15.76$

Then divide by 7.5 = 15.76/7.5 = ans 2 sig.